

Homework #1

Due date: **28/10/2022**

Notes:

- Your work (code + written answers) must be submitted through SUCourse+.
- Winzip your programs and add a readme.txt document to explain the programs and how to use them.
- Name your winzip file as **"cs411_507_hw01_yourname.zip"**

1. (15 pts) Consider the shift cipher. Show that the ciphertext "NGZZK" can be decrypted into two meaningful English words. Find out those words and corresponding encryption keys.

BUNNY $\Rightarrow$ NGZZK (key = 12)

SLEEP $\Rightarrow$ NGZZK (key = 21)

2. (20 pts) Consider the ciphertext generated by Affine Cipher over Z_{26} . As a hint, you are told that the most frequent letter in the plaintext is 'T'. Find the plaintext, the encryption and decryption keys. Show your work.

"ZJOWMJ ZJGC BS UEVRSCC, KSZ ZJSFS GC USZJOV GR GZ."

Encryption keys: (23, 4) and Decryption keys: (17, 10)

The plaintext: "THOUGH THIS BE MADNESS, YET THERE IS METHOD IN IT."

3. (15 pts) Assume that you design a new affine cipher where you encrypt two letters at a time, where your alphabet is

{'A':0, 'B':1, 'C':2, 'D':3, 'E':4, 'F':5, 'G':6, 'H':7, 'I':8, 'J':9, 'K':10, 'L':11, 'M':12, 'N':13, 'O':14, 'P':15, 'Q':16, 'R':17, 'S':18, 'T':19, 'U':20, 'V':21, 'W':22, 'X':23, 'Y':24, 'Z':25, ' ':26, ' ':27}.

In other words, you group your plaintext message in bigrams (i.e., two-character words) and encrypt each bigram of the plaintext separately using this affine cipher. For example, if the first two letters of a plaintext is "TH" then it will be encoded as follows

$$TH \Rightarrow 19 \times 28 + 7 = 539.$$

If the number of letters in the plaintext is not a multiple of two, you pad it with the letter "X" at the end. Determine the modulus and the size of the key space.

Theoretically the number of bigrams are $28 \times 28 = 784$

Then the modulus is 784

The size of the key space: $\phi(784) \cdot 784 = 336 \cdot 784 = 263424$

4. (15 pts) Is the affine cipher defined in question (3) secure against the letter frequency analysis?

It is better than the original affine cipher. However, the language statistics is not totally removed. Note the following trigram frequencies of the English language.

TH : 2.71	EN : 1.13	NG : 0.89
HE : 2.33	AT : 1.12	AL : 0.88
IN : 2.03	ED : 1.08	IT : 0.88
ER : 1.78	ND : 1.07	AS : 0.87
AN : 1.61	TO : 1.07	IS : 0.86
RE : 1.41	OR : 1.06	HA : 0.83
ES : 1.32	EA : 1.00	ET : 0.76
ON : 1.32	TI : 0.99	SE : 0.73
ST : 1.25	AR : 0.98	OU : 0.72
NT : 1.17	TE : 0.98	OF : 0.71

Check also the following link:

<http://practicalcryptography.com/cryptanalysis/letter-frequencies-various-languages/english-letter-frequencies/>

5. (20 pts) Consider the following ciphertext that is encrypted with the affine cipher defined in question (3):

"ZDZUKEO.AANDOGIJTLNEKEPHZUQDX NDS VLNDJGQLYDVSBU.DER.K.UYT"

Find the key and decrypt the ciphertext.

(Hint 1: The plaintext is a sentence that ends with a dot.)

(Hint 2: The length of the plaintext (plen) is not a multiple of 2; $\text{plen} = 2k+1$ for an integer k.)

Encryption key: (185, 524)

Decryption key: (89, 404)

"I HAVE COME TO BELIEVE THAT THE WHOLE WORLD IS AN ENIGMA."

6. (15 pts) If we select a different shift amount for every letter in the plaintext uniformly randomly, the shift cipher becomes a one-time-pad with perfect security.

Suppose p_α is the probability of the plaintext letter α from the Turkish alphabet, where $\alpha \in \{A, B, C, \zeta, \dots, Z\}$. Suppose also that p_β is the probability of the ciphertext letter β , where $\beta \in \{A, B, C, \zeta, \dots, Z\}$. Demonstrate that $p_\beta = 1/29$ for every $\beta \in \{A, B, C, \zeta, \dots, Z\}$ independent of the values of p_α .

Assume that the ciphertext letter β encodes to $i \in \{0, 1, \dots, 28\}$. There are 29 different combinations of the shift amount and plaintext letter that will encrypt to β :

$$j + k \equiv i \pmod{29}$$

where j encodes the plaintext letter and $j = 0, 1, \dots, 28$. Then p_β can be written as

$$p_\beta = (1/29)p_A + (1/29)p_B + \dots + (1/29)p_Z = (1/29)(p_A + p_B + \dots + p_Z) = 1/29$$

7. BONUS (20 pts) The following was encrypted using the Vigenere cipher:

“JR WYDUGQ AR LRG BTFWB’U UECDC YVTF S CYVNE LY JVS QZYWYDCJC, CAD FAC
NRGQ KZTRAB MXYVTRAXIYY, YK SH GHC DOXRL DDYQES UWBG GIJLSPT UN SXF
FILCSPT DMOX VB TFW RGNVC SXF YULYO QS TFW CGN. TFW GKQE PGYOF SCWWGQ
TMG XCERMO PQE HGK BQYLGfQ INIR, SXF GO FAWURLD ZO YNS GF DGERMJ VGFT
FAC DEOYV CJBUIVOTF SFGENQ CMDVKQE UADJ GHC VYQEWYQC QE SUWOR GHC
TBKP-A-ZJKE SRME DJR LMO WCATCD. RG EEAGSNRD DJYO FIBW DQ FIBW
LGGWCWX VUE TSBKBUQ GLLRCRK KPQ MSDDKCLGWN VUE FSJCEDQ LRCG IL
JOCYIRQ VQQGCV YPYY GF RKF MGFN. DRTUWOP N GPSXF CIYFY CAD Y
UOPGRC-LKDYE NAVGQ HGYR YVTF TYQXS USC UCAAW PQE A FSVH N DMROP GO
USVM NBPWKUG, YCL RG RSQSIGQ IR OSVU TPWZKQARAYP. UIQ ZOCYI YJWU UULY
VQBSCDI CG HGK CKQEQ. ZO FVD LGD MAOU ORCG TM VY YVTF LRQFE YJWU NNB
ZKPQS, YFN YUEL, LY JVS CPMKGEB NSUVOL, GXG NRK KOGZEB DSCOLC LY DEUQZ
KINILKD VUE ZGYMF OL LRG GAZDO, JR LSJMJRD YOKA YIIW K HEIEZDGAEB ZYTFE,
ZSBGYY KACUVNE LRG CIYFY UGOMD. RG JARURGQ TFW OCFY USVM BF RZO QGHCI
SP SRMFD QS HGE, KPQ FMJ DJR FGJCV GIKW BGNLGROF GHYL RKF WYDU YNS
BAPHRRCFD HEOK LRCG OD GDJRR KWX. JR EVHOTVELUOF N MMEOPGAPQ ZCAG
MX CJNMC LRCG HC KRQHLB OKNX SM MXEBURZVA. GHC KGGNT ZMBUG TFJYWITH
RZO UXIL GP JVS DGBGUEYV SP GILQ LGNDQ, SXF UE NSEURD YFN OBPNNWN JVS
ZJYPMEB XKER WGLR JVS FSXFXEPURKRF.”

Attack it and find the key length and the key. Note that only the letter characters are encrypted.

By following the algorithm steps mentioned in the lectures, you will be able to reach the following key and plaintext values.

KEY: CNAYSK

The Plaintext:

HE WALKED AT THE OTHER'S HEELS WITH A SWING TO HIS SHOULDERS, AND HIS LEGS SPREAD UNWITTINGLY, AS IF THE LEVEL FLOORS WERE TILTING UP AND SINKING DOWN TO THE HEAVE AND LUNGE OF THE SEA. THE WIDE ROOMS SEEMED TOO NARROW FOR HIS ROLLING GAIT, AND TO HIMSELF HE WAS IN TERROR LEST HIS BROAD SHOULDERS SHOULD COLLIDE WITH THE DOORWAYS OR SWEEP THE BRIC-A-BRAC FROM THE LOW MANTEL. HE RECOILED FROM SIDE TO SIDE BETWEEN THE VARIOUS OBJECTS AND MULTIPLIED THE HAZARDS THAT IN REALITY LODGED ONLY IN HIS MIND. BETWEEN A GRAND PIANO AND A CENTRE-TABLE PILED HIGH WITH BOOKS WAS SPACE FOR A HALF A DOZEN TO WALK ABREAST, YET HE ESSAYED IT WITH TREPIDATION. HIS HEAVY ARMS HUNG LOOSELY AT HIS SIDES. HE DID NOT KNOW WHAT TO DO WITH THOSE ARMS AND HANDS, AND WHEN, TO HIS EXCITED VISION, ONE ARM SEEMED LIABLE TO BRUSH AGAINST THE BOOKS ON THE TABLE, HE LURCHED AWAY LIKE A FRIGHTENED HORSE, BARELY MISSING THE PIANO STOOL. HE WATCHED THE EASY WALK OF THE OTHER IN FRONT OF HIM, AND FOR THE FIRST TIME REALIZED THAT HIS WALK WAS DIFFERENT FROM THAT OF OTHER MEN. HE EXPERIENCED A MOMENTARY PANG OF SHAME THAT HE SHOULD WALK SO UNCOUTHLY. THE SWEAT BURST THROUGH THE SKIN OF HIS FOREHEAD IN TINY BEADS, AND HE PAUSED AND MOPPED HIS BRONZED FACE WITH HIS HANDKERCHIEF.